School ID

AUSTRALIAN SCIENCE INNOVATIONS

Australian Science Olympiads

2020 AUSTRALIAN SCIENCE OLYMPIAD EXAM

BIOLOGY

TO BE COMPLETED BY THE STUDENT. USE CAPITAL LETTERS.

Student Name: ……………………………………………………………………………….

Home Address: ……………………………………………………………………………..

……………………………………………………………….. **Post Code:** …………...

Telephone: (……….) ………………………………. **Mobile:** ………………………….

E-Mail: ………………………………………………………... **Date of Birth:** ……/……/……

□ Male □ Female □ Unspecified Year 10 □ Year 11 □ Other: …….

Name of School: ………………………………………………………….**State:** ……...

To be eligible for selection for the Australian Science Olympiad Summer School, students must be an Australian citizen at the time of offer.

The Australian Olympiad teams in Biology, Chemistry, Earth and Environmental Science and Physics will be selected from students participating in the Science Summer School.

Please note - students in Year 12 in 2020 are not eligible to attend the 2020 Australian Science Olympiad Summer School.

□ I am an Australian public high school student and would like to be considered for the Australian Science Olympiad Summer School Scholarship.

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Examiners Use Only:

2020 AUSTRALIAN SCIENCE OLYMPIAD EXAM

BIOLOGY

Time Allowed

Reading Time: 10 minutes

INSTRUCTIONS

- ***Attempt all questions in ALL sections of this paper.***
- **Permitted materials: non-programmable, non-graphical calculator, pens, pencils, erasers and a ruler.**
- **Answer all questions on the MULTIPLE CHOICE ANSWER SHEET PROVIDED. Use a pencil.**
- **Marks will not be deducted for incorrect answers.**

Examination Time: 120 minutes

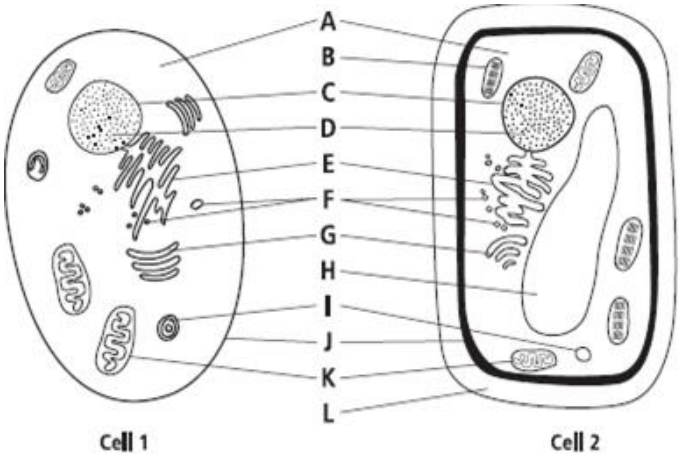
MARKS

- **1 mark for each question**
- **Total marks for the paper 60 marks**

Integrity of Competition

If there is evidence of collusion or other academic dishonesty, students will be disqualified. Markers' decisions are final.

Questions 1 – 3 relate to the following image of two cells.



1. Cell 1 is best described as what type of cell?

    a. red blood cell

    b. prokaryotic cell

    c. plant cell

    d. eukaryotic cell

2. What does the organelle **K** produce when functioning normally in a cell?

    a. protein

    b. mRNA

    c. ATP

    d. endoplasmic reticulum

3. Label **B**, **H** and **L** are unique to cell 2. These labels represent what organelles

    a. mitochondria, vacuole, cell membrane

    b. chloroplast, vacuole, cell wall

    c. endoplasmic reticulum, nucleus, cell wall

    d. chloroplast, vacuole, cell membrane

4. What is the term that best describes the net movement of uncharged molecules from an area of high concentration to an area of low concentration?

   a. diffusion

   b. active transport

   c. passive transport

   d. osmosis

Questions 5 and 6 relate to the following information

A group of scientists measured the distance covered over 24 hours by two samples of a common periwinkle *(Littorina littorea)*

Individual	**Sample A (cm)**	**Sample B (cm)**
1	12	30
2	13	9
3	20	17
4	3	11
5	11	18
6	21	5
7	18	6
8	22	13
9	8	11
10	0	19
Standard Deviation	**7.58**	**7.42**

The mean of a sample is defined as

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

The mean of sample A is 12.8cm

The *skew* of a sample is defined as

$$skew = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s}\right)^3$$

where s represents standard deviation

5. What is the skew of sample A?

   a. 0.000

   b. -0.4176

   c. 1.0353

   d. -0.0241

6. Which of the following statements is **true**?

   a. The mean of sample B is greater than the mean of sample A

   b. The skew of sample B is greater than the skew of sample A

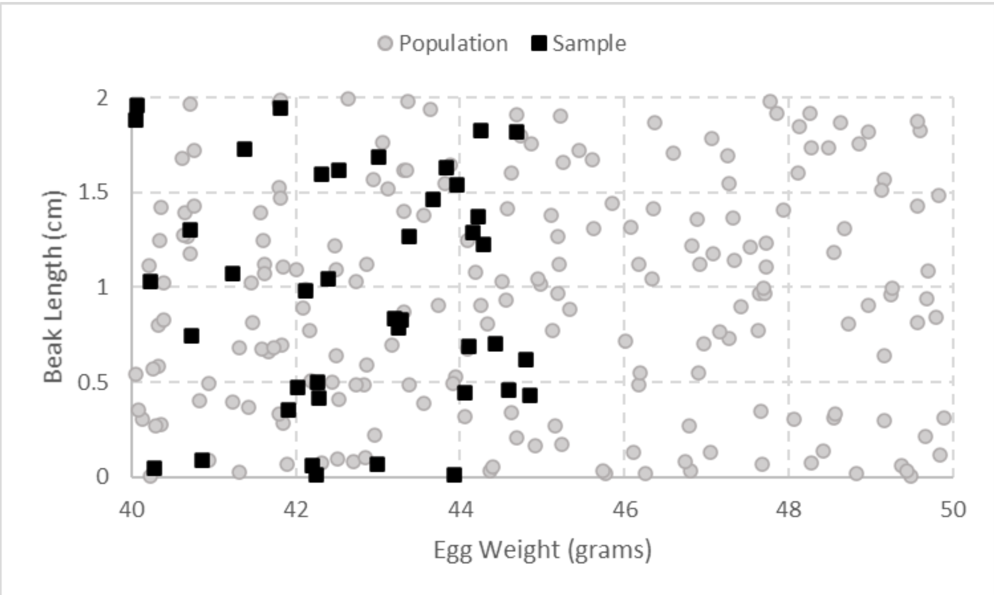
   c. The data in sample A is grouped more closely around the mean than the data in sample B

   d. Sample B contains more observations than sample A

Questions 7 and 8 relate to the following information

Anna is studying a population of seabirds which roost on one particular island off the coast of Tasmania. She hypothesises that seabirds with longer beaks lay heavier eggs.

Anna recorded the beak length and egg weight of a sample of 40 birds. Her results and the population of seabirds are plotted below



7. Which of the following is supported by Anna's data?

   a. The population has average egg weight of roughly 43 grams

   b. Birds with longer beaks tend to lay heavier eggs

   c. Birds with longer beaks tend to lay lighter eggs

   d. Birds with longer beaks have a competitive advantage

8. Which of the following is **true**?

   a. The difference between Anna's sample and the population could be attributed to random error

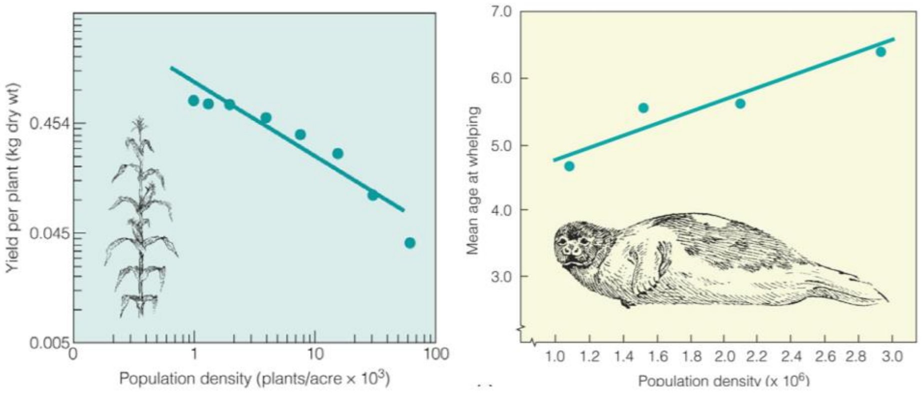
   b. There is a clear relationship between egg weight and beak length in the population

   c. Anna's sample overestimates the average egg weight of the population

   d. Increasing the sample size would not improve Anna's study

Questions 9, 10 and 11 relate to the following information

On the left is corn yield, on the right is seal reproduction. Interpret these and answer the questions below:



9. In the corn figure, the declining line indicates:

   a. the influence of global warming on corn crops.

   b. an increase in available resources as population density increases.

   c. increasing competition as population density increases

   d. decreasing competition as population density increases

10. In the seal figure, the positive line may suggest:

  a. more food available as the seals age

  b. as the population density increases, the seals have less food

  c. as the population density decreases, seal age at reproduction increases.

  d. less food available as the seals age

11. These two figures are evidence of:

  a. density dependent competition

  b. keystone predators

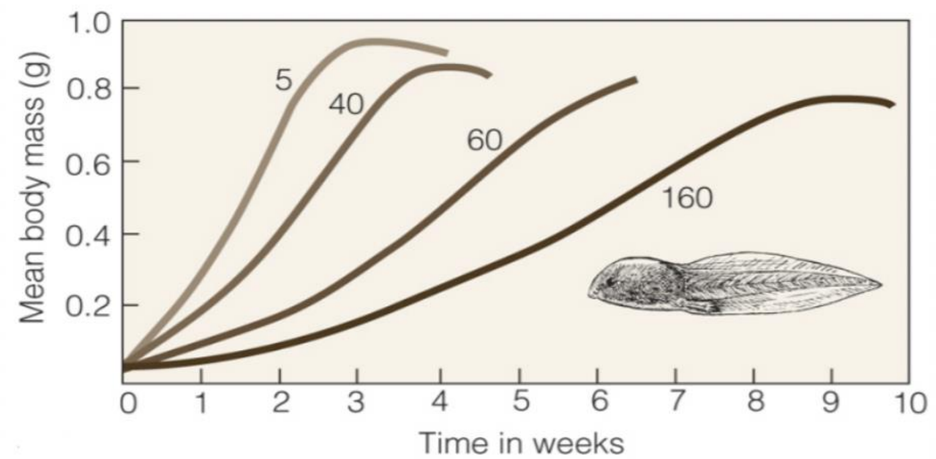
  c. density independent population regulation

  d. resource scarcity increases as the density of a population decreases

Questions 12, 13 and 14 relate to the following information

The following graph depicts the results of a study where the number of tadpoles within a particular habitat (finite resources) were experimentally varied.

Note that the numbers next to the lines are the numbers of individuals in each treatment.



12. As the number of tadpoles in the habitat increases, the mean body mass at week three:

  a. increases

  b. decreases

  c. stays the same

  d. cannot tell from this diagram

13. This figure indicates that:

a. access to resources decreases as population size decreases

b. access to resources decreases as population size increases

c. access to resources is not affected by population size

d. carrying capacity does not affect population size

14. This figure provides support for:

a. density-dependent population regulation

b. density-independent population regulation

c. neither of these density population effects

d. both of these density population effects

Questions 15, 16 and 17 relate to the following information

Rhizophores are leafless branches that arise from the stem and grow downward, producing roots at its tip when it reaches the soil. Rhizophores are commonly known as aerial roots. Scientists measured length and height of rhizophores of mangrove plant (*Rhizophora mangle*, Fig A). They also made cross sections of rhizophores and observed their anatomical characteristics. The rhizophore cross sections were classified in orders according to the number of arches away from the main stem. The results are shown in Fig B and Fig C.

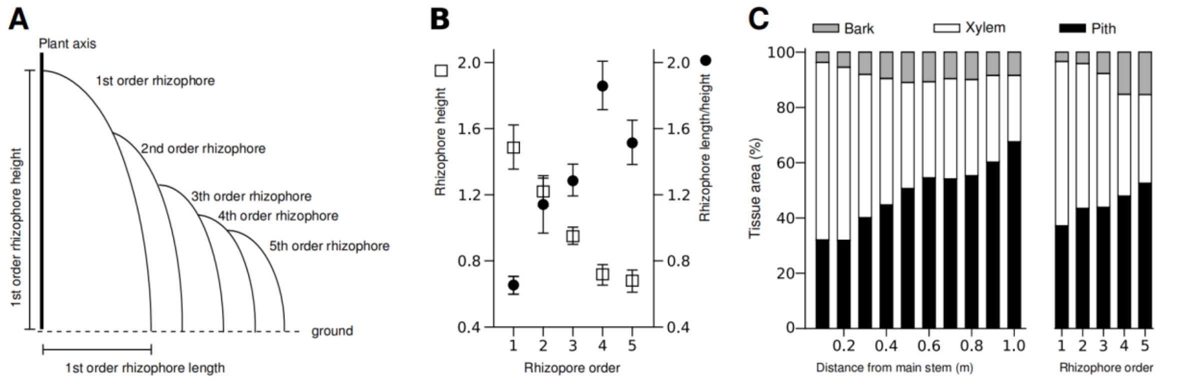


Fig: Rhizophores of *Rhizophora mangle* plants.

A: Rhizophore height and length measurement.

B: Change in height (empty square) and in length/height proportion (full circle) in five sequential orders of rhizophores.

C: Relative proportions of bark (including aerenchyma), xylem and pith along the length of individual first-order rhizophores (left), and at the base of rhizophores of sequential orders (right).

15. What mechanism makes water rise from the root system to the leaves

a. forces created in the cells of the root

b. increased respiratory activity in root cells

c. osmotic force in the shoot system

d. tension in the cell sap due to transpiration

16. Which of the following statements is correct?

a. Rhizophore height and the length/height proportion in the rhizophores decreases with rhizophore order.

b. Within first-order rhizophores, the xylem proportion in the cross-section is larger when closer to the main stem, and decreases progressively as the rhizophore approaches the ground.

c. When rhizophore order changes from 1 to 5, bark and pith proportion decreases, while xylem proportion increases.

d. The supportive function is likely enhanced in the first-order rhizophores, with lower length/height proportion, and the lowest proportion of xylem.

17. How does water support a mangrove plant?

a. By transpiration, the process whereby the water is lost at the leaves, enabling constant cycling of water throughout the plant.

b. By making the cells flaccid, enabling the mangrove plant to strengthen its structural integrity.

c. By filling the stem with water, assisting the stem to resist gravitational as well as osmotic forces.

d. By making the cells turgid, promoting rigidity of plant structures.

Questions 18 and 19 relate to the following information

Sucrose is produced in leaves and translocated short and long distances through veins to non-photosynthetic organs such as roots, stems, flowers and fruits. Sucrose transport is essential for the distribution of carbohydrates in plants. Two principal pathways include symplast and apoplast by which sucrose molecules are transported in the phloem of leaves as shown in the Figure. In the symplastic pathway, sucrose moves from cell to cell via plasmodesmata, which are continuous junctions between neighbouring plant cells. The apoplastic pathway involves a step where sucrose is momentarily transported outside of a plant cell plasma membrane.

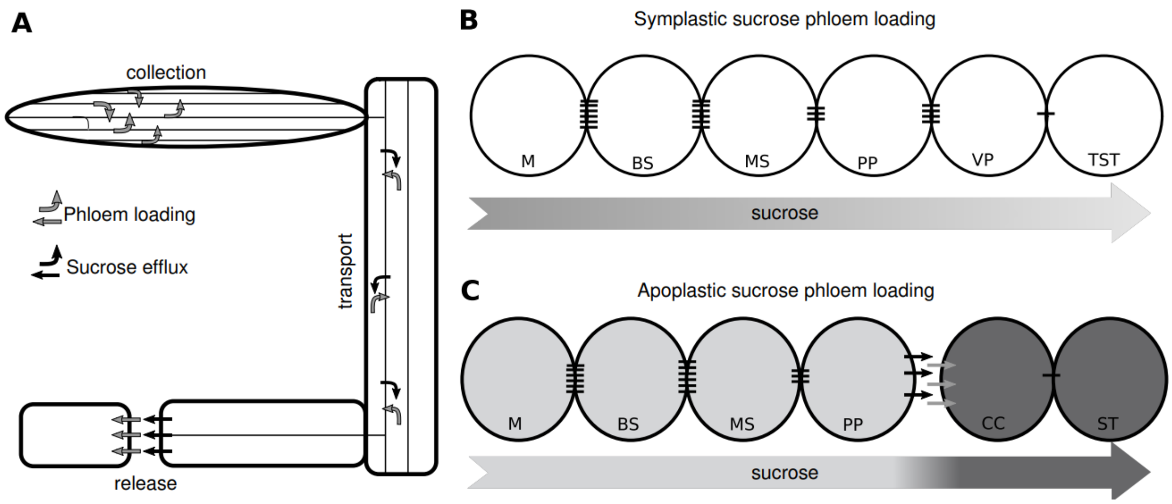


Figure. Diagram of the whole plant phloem network. M - Mesophyll, BS - Bundle sheath, MS - Mestome sheath, PP - Phloem parenchyma, VP - Vascular parenchyma, CC - Companion cell, TST - Thick walled sieve element, ST - Sieve element.

18. Which statement about phloem is true? It carries:

    a. nitrate ions from the leaves to the roots

    b. sugars from the roots to the leaves

    c. nutrients to growing regions of the plant

    d. water from the roots to the leaves

19. Which of the following is correct?

a. Sucrose is synthesised in leaves and transported long distance through phloem to sinks under hydrostatic pressure gradient.

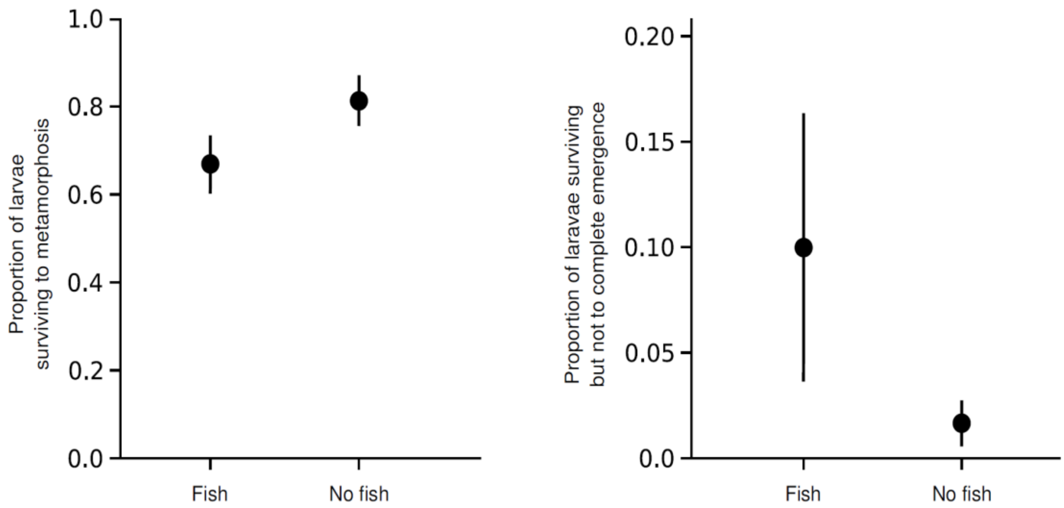
b. Loading sucrose in the symplastic pathway requires energy at several steps due to the movement across the secondary walls of living cells.

c. In the apoplastic pathway, sucrose molecules are passively loaded through plasmodemata.

d. Unloading sucrose molecules at the sinks requires no energy release because of movement against a gradient concentration of sucrose.

Questions 20 relates to the following information

A population of dragonfly larvae (*Leucorrhinia intacta*) is separated into two groups. In both groups, larvae populations are put inside a cage with no food limitation. The first group is exposed to a fish predator that can swim freely, but cannot enter the cage. The second group is a control with no fish. The proportion of larvae surviving and the proportion of live larvae failing to metamorphose in the two groups are shown below:



20. Which of the following is correct?

a. One of the causes of high failure rate of metamorphosis of the larvae upon exposure to a non-lethal predator is cannibalism.

b. The high mortality of larvae in the first group is due to predator-induced stress.

c. In the predator treatment, the percentage of individuals that survived the larval stage completed emergence to the adult stage is higher than the percentage of those in the fishless treatment.

d. The survival of dragonfly before metamorphosis is dependent on the predator while those during metamorphosis is not.

Questions 21 relates to the following information

Male guppies (*Poecilia reticulata*) show a complex colour pattern polymorphism that varies with predation pressure, reflecting a balance between selection for camouflage by predators and selection for attention by sexual selection. Three experimental ponds were used to study this phenomenon, mimicking the real condition on the native habitat of the guppies and its predators, *Rivulus hartii* and *Crenicichla alta*. One pond has the control group while the other two ponds were had one of the two predators added. In the field, *C. alta* was observed to be more dangerous than *R. hartii*.

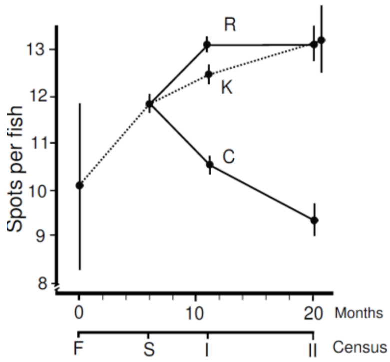


Figure: Changes in the number of spots per fish during the course of the experiment. Line 'K' stands for pond without predator, 'R' for pond with *R. hartii*, and 'C' for pond with *C. alta*. In the X-axis, 'F' stands for the time when the foundation population was started, 'S' stands for the time when the predators were added, then 'I' and 'II' stands for the numbering of the following censuses.

21. Which of the following is correct?

    a. The colour pattern is responsible for the reduced fitness of *P. reticulata*.

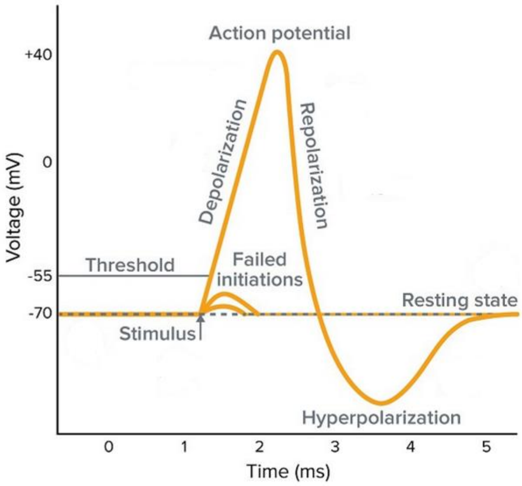
    b. Sexual selection for colour pattern of the *P. reticulata* cannot be inferred from the data.

    c. The colour pattern of the *P. reticulata* may be advantageous in escaping *R. hartii*.

    d. The two predators possibly use two different mechanisms to detect *P. reticulata*.

Questions 22, 23 and 24 relates to the following information

The following graph illustrates the change in membrane potential (mV) during an action potential.



22. During the resting state, the membrane potential is -70mV. Which of the following statements is true?

    a. In the resting state, there is no movement of ions across the plasma membrane.

    b. The membrane potential is maintained by the balance between active transport and simple diffusion of ions across the membrane.

    c. In the resting state, the concentration of potassium ions is greater inside the cell than outside the cell.

    d. If the cell is not in the resting state, an action potential cannot be generated.

23. Changes in the membrane potential that do not reach the threshold voltage are known as graded potentials. Which of the following statements is false?

    a. The size of a graded potential is proportional to the stimulus, whereas the size of an action potential is the same regardless of the stimulus size.

    b. Graded potentials may be depolarising or hyperpolarising, whereas all action potentials are depolarising.

    c. Graded potentials propagate bidirectionally in a neuron, whereas action potentials are unidirectional.

    d. Both graded potentials and action potentials result in the opening of voltage-gated ion channels.

24. A drug that prevents the opening of voltage-gated sodium channels is administered. What effect will this have on the action potential?

  a. Repolarisation will occur more slowly.

  b. The threshold voltage required to produce an action potential will increase.

  c. No action potential will be generated, regardless of the size of the stimulus.

  d. Voltage-gated potassium channels will open instead of voltage-gated sodium channels.

Questions 25 relates to the following information

Myasthenia gravis is a chronic autoimmune condition that affects the junction between neurons and muscles, causing serious muscle weakness. In most cases, it is caused by the production of antibodies against acetylcholine neurotransmitter receptors on the muscle.

25. Which of the following options is least likely to be used as a treatment for people with myasthenia gravis?

  a. Anti-acetylcholinesterase agents to prevent the breakdown of acetylcholine in the neuromuscular junction.

  b. Immunosuppressants to reduce the activity of the immune system.

  c. Surgical removal of the thymus to prevent antibody production.

  d. A high sodium diet to increase the strength of action potentials.

Questions 26 and 27 relates to the following information

The following table provides information about three different solutions, with respect to the plasma membrane.

Solution	Diffusible across plasma membrane
0.9% sodium chloride solution	No
5.0% dextrose (glucose) solution	No
1.8% urea solution	Yes

These solutions can all be treated as iso-osmolar to human cells.

26. Which of the following is not a characteristic of the plasma membrane?

    a. It is selectively permeable.

    b. Cholesterol helps to maintain membrane fluidity at low temperatures.

    c. It is composed of a phospholipid bilayer.

    d. Peripheral proteins enable facilitated diffusion.

27. Which of the following statements about the three solutions is true?

    a. Only 1.8% urea solution is isotonic to human cells.

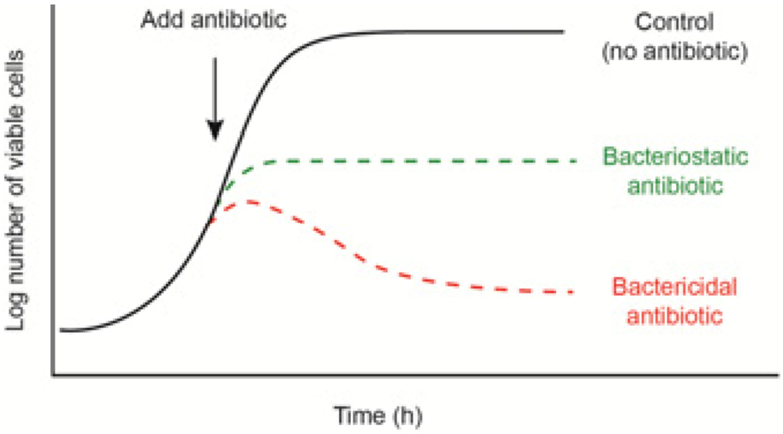
    b. Only 0.9% saline and 5.0% dextrose are isotonic to human cells.

    c. Only 1.8% urea and 0.9% saline are isotonic to human cells.

    d. All three solutions are isotonic to human cells.

Question 28 relates to the following information

The following graph illustrates the antimicrobial effect of antibiotics in vitro.



28. Which of the following statements is supported?

    a. In people who are immunocompromised, bacteriostatic drugs are less effective than bactericidal drugs.

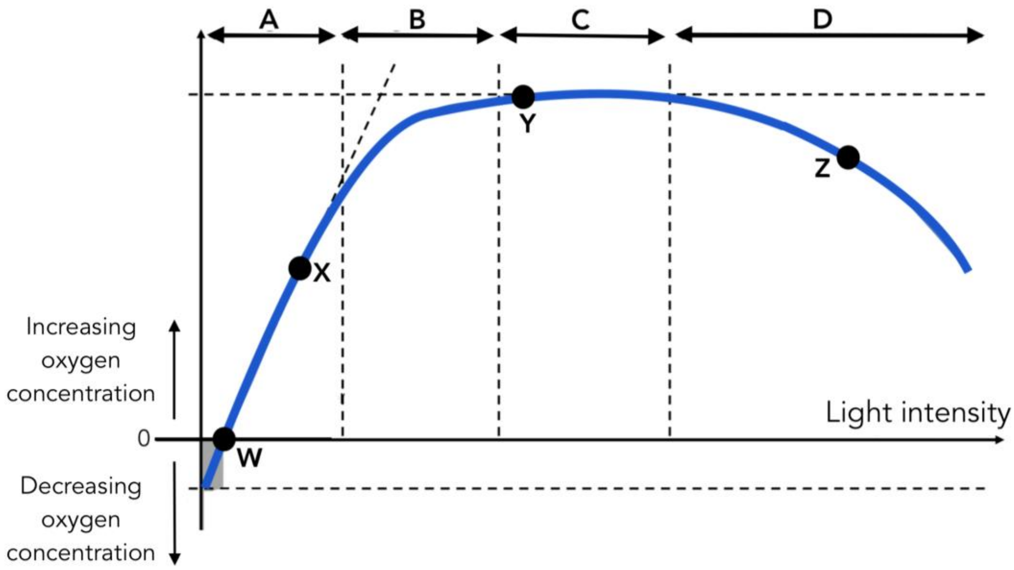
    b. When given to treat bacterial infection, bactericidal drugs lead to a quicker recovery than bacteriostatic drugs.

    c. Bactericidal drugs tend to cause more adverse effects than bacteriostatic drugs.

    d. If a patient is only given bacteriostatic drugs for a bacterial infection, they will never be cured of the infection.

Questions 29, 30 and 31 refer to the following information

The following graph shows relative oxygen concentrations in an airtight container containing one spinach plant exposed to varying light intensities. The y axis indicates oxygen concentration and the x axis indicates increasing light intensity from left to right. The units of both axes are arbitrary; they do not matter.



29. Consider the stages A, B, C and D marked at the top of the graph. At which stage is light limiting the rate of photosynthesis?

    a. A

    b. B

    c. C

    d. D

30. Consider point X labelled on the graph. Select the CORRECT option:

    a. At point X, the concentration of carbon dioxide is increasing.

    b. At point X, the concentration of carbon dioxide is decreasing.

    c. At point X, the concentration of carbon dioxide is constant.

    d. Insufficient information to answer the question.

31. Consider the points on the graph labelled W, X, Y and Z. Each point has been paired up with a description of what is happening at that point. Select the INCORRECT pairing.

a. At point W, the rate of CO_2 consumption balances the rate of CO_2 production.

b. At point X, the rate of photosynthesis exceeds the rate of respiration.

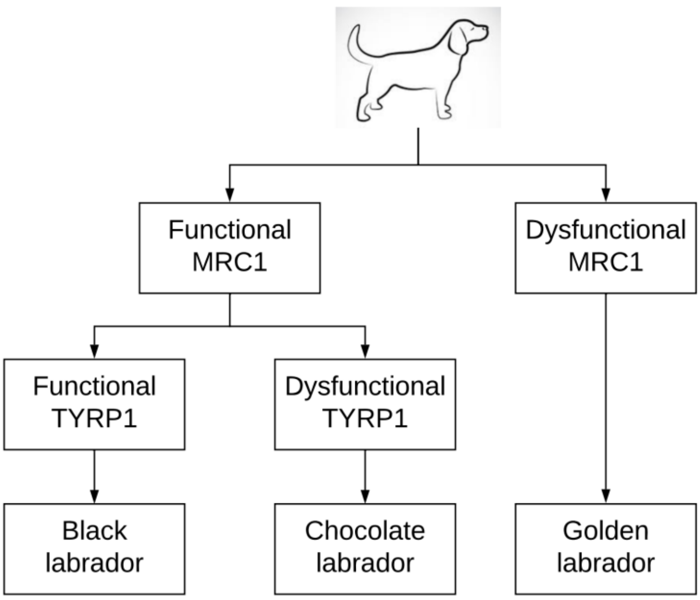
c. At point Y, CO_2 concentration is a possible limiting factor.

d. At point Z, O_2 concentration is decreasing and therefore the rate of respiration is higher than the rate of photosynthesis.

Questions 32, 33, 34 and 35 refer to the following information

The three recognised colours of Labradors are black, chocolate and golden. These coat colours result from differences in two genetic loci which have alleles B/b and E/e respectively.

Supplied is a simplified flowchart representing pigment deposition in labradors. E is the allele for functional MRC1 protein, while e is a mutated allele resulting in no MRC1 protein. B is the allele for functional TYRP1 enzyme, while b is a mutated allele resulting in no TYRP1 enzyme activity. If there is no pigment deposition, the labrador is considered golden.



32. Which one of the following could be the genotype of a black female Labrador?

a. bbEE

b. BBee

c. BbEe

d. Bbee

33. If a double heterozygote female black Labrador is mated with a male Labrador of the same genotype, what is the phenotypic ratio of their offspring expected to be?

a. 9 black: 3 chocolate: 4 golden

b. 9 black: 6 chocolate: 1 golden

c. 12 black: 3 chocolate: 1 golden

d. 1 black: 2 chocolate: 1 golden

34. In a cross of a chocolate female with a chocolate male, which phenotypic ratio is IMPOSSIBLE?

a. All chocolate

b. 3 chocolate to 1 golden

c. 1 black to 1 chocolate

35. When a golden female Labrador was mated with a chocolate male, half of the puppies were chocolate and half were golden. What is the genotype of this golden female?

a. BBee

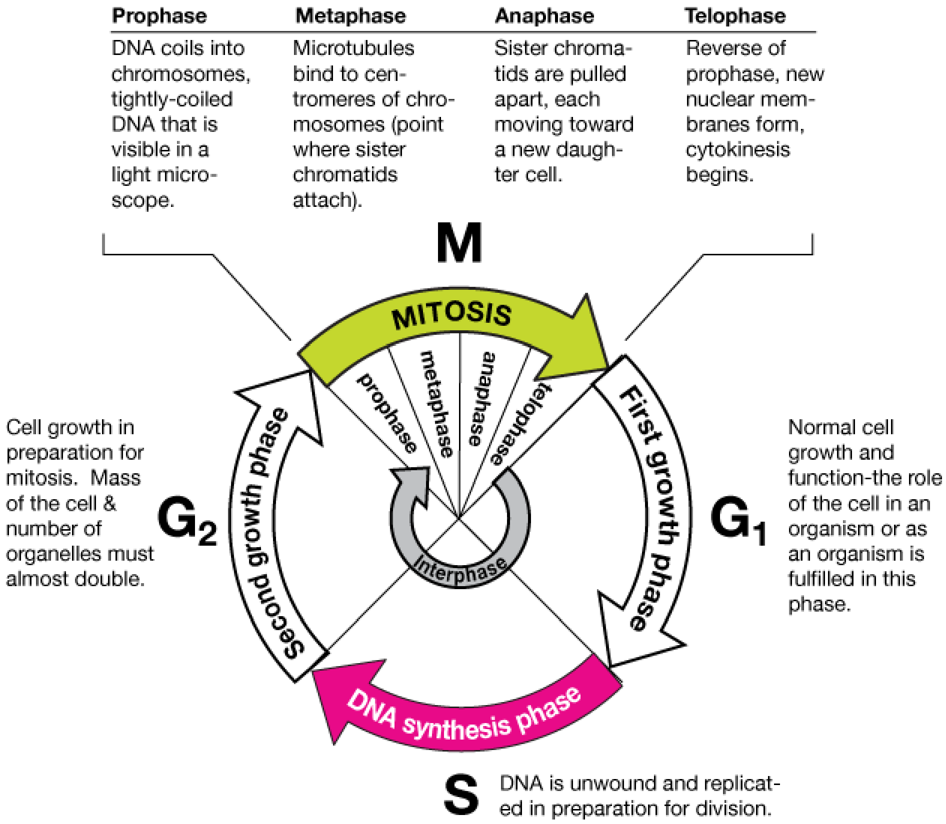
b. Bbee

c. bbee

d. bbEe

Questions 36, 37 and 38 refer to the following information

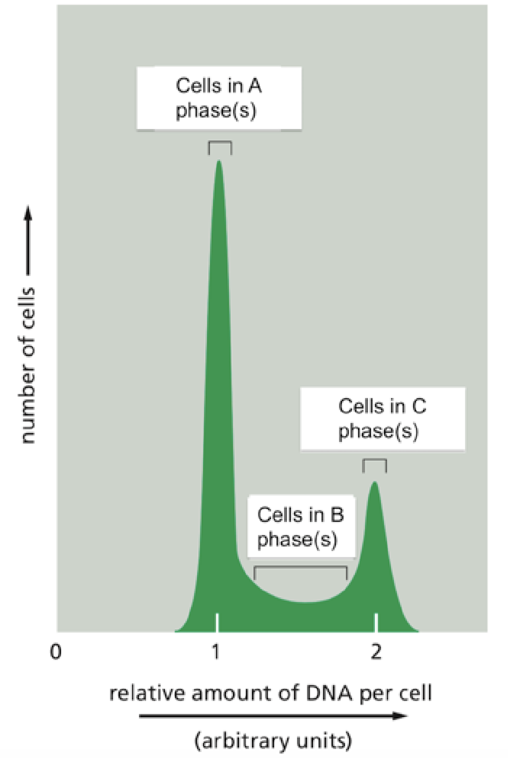
This diagram summarises the phases of the cell cycle.



A sample of somatic cells in various stages of the cell cycle are labelled with a fluorescent dye that exclusively binds to DNA. A technique called flow cytometry is then used to:

1. assess the intensity of the fluorescence in each single cell,
2. sort cells with different fluorescent intensities into collection tubes, and
3. count the number of cells in each collection tube.

The following graph is generated.



36. Identify cell cycle phase(s) A, B and C on the diagram above.

	Phase(s) A of cell cycle	Phase(s) B of cell cycle	Phase(s) C of cell cycle
A	Telophase, cytokinesis, G1	S and anaphase	Interphase, prophase, metaphase, G2
B	G1	S	G2 and M
C	Telophase, cytokinesis	Anaphase	Interphase, prophase, metaphase
D	M	S and G2	G1

37. What is the LEAST plausible reason as to why some cells may display a relative amount of DNA less than 1 arbitrary unit?

   a. Some cells may be dead, and their DNA may have degraded over time.

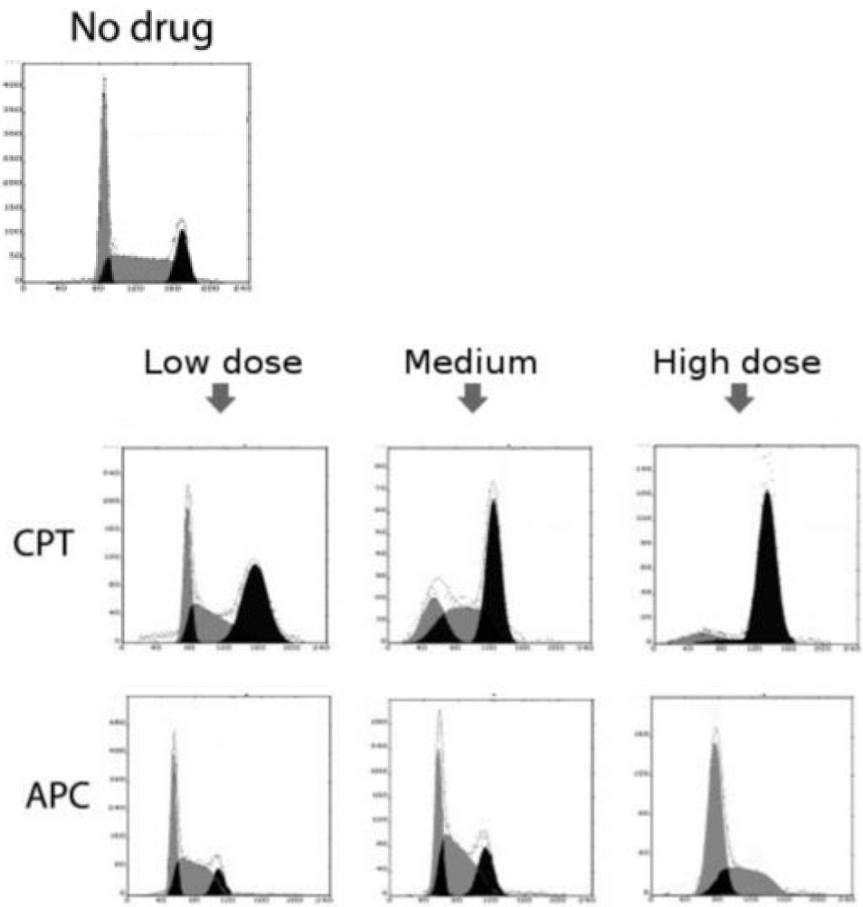
   b. The measurement of fluorescence may not be completely exact, leading to random fluctuations.

   c. The fluorescent dye may not bind to DNA perfectly.

   d. They are haploid cells lacking multiple chromosomes due to non-disjunction.

This additional information relates to Question 38.

The following graphs are generated from flow cytometry analysis. The y-axis denotes cell count and the x-axis represents increasing DNA content from left to right. CPT (camptothecin) and APC (aphidicolin) are hypothesised to inhibit the cell cycle at various stages. One sample of cells is left untreated (no drug) as a control.



38. Interpret the graphs to identify the phase in which cell cycle arrest has occurred for both drugs.

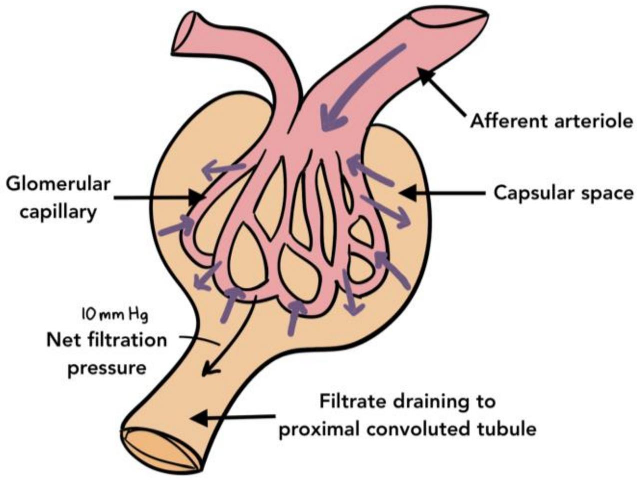
	CPT caused cell cycle arrest in…	APC caused cell cycle arrest in…
A	G2/M phase	G1 phase
B	G1 phase	S phase
C	G2/M phase	No effect (APC is not a cell cycle inhibitor)
D	G1 phase	G1/S phase

Questions 39 and 40 refer to the following information.

Every day, some 200 litres of blood are filtered by your kidneys. Most of that fluid is reabsorbed back into the body; the rest of it becomes urine. The basic unit of structure in the kidney is the nephron. Blood enters the nephron through the afferent arteriole which branches into a network of glomerular capillaries. A series of hydrostatic and osmotic pressures then draw fluid in or drive fluid out of these capillaries. The filtrate is collected in the capsular space which drains into the proximal convoluted tubule (PCT) of the nephron. This is the basis of renal filtration.

Glomerular hydrostatic pressure (GHP) is the pressure exerted by blood on the walls of the glomerular capillaries. Capsular hydrostatic pressure (CHP) is the pressure exerted by the filtrate in the capsular space. Blood colloid osmotic pressure (BCOP) is the pressure exerted by the concentration of proteins in the blood. No proteins are expected to found in the capsular space as the glomerular capillary walls do not allow them to pass. The net filtration pressure (NFP) is the sum of all these pressures drawing fluids in or out of the glomerular capillaries.

The arrows in the diagram represent fluid moving in and out of the capillaries under the influence of these pressures.



39. In which direction would blood colloid osmotic pressure (BCOP) drive fluid?

    a. From the glomerular capillaries into the capsular space.

    b. From the capsular space back into the glomerular capillaries.

    c. BCOP does not have an effect on the direction of fluid movement because no proteins are expected to be found in the capsular space.

    d. Insufficient information to tell.

40. Which option expresses the correct relationship between these four pressures?

a. NFP = CHP + BCOP - GHP

b. NFP = CHP - BCOP - GHP

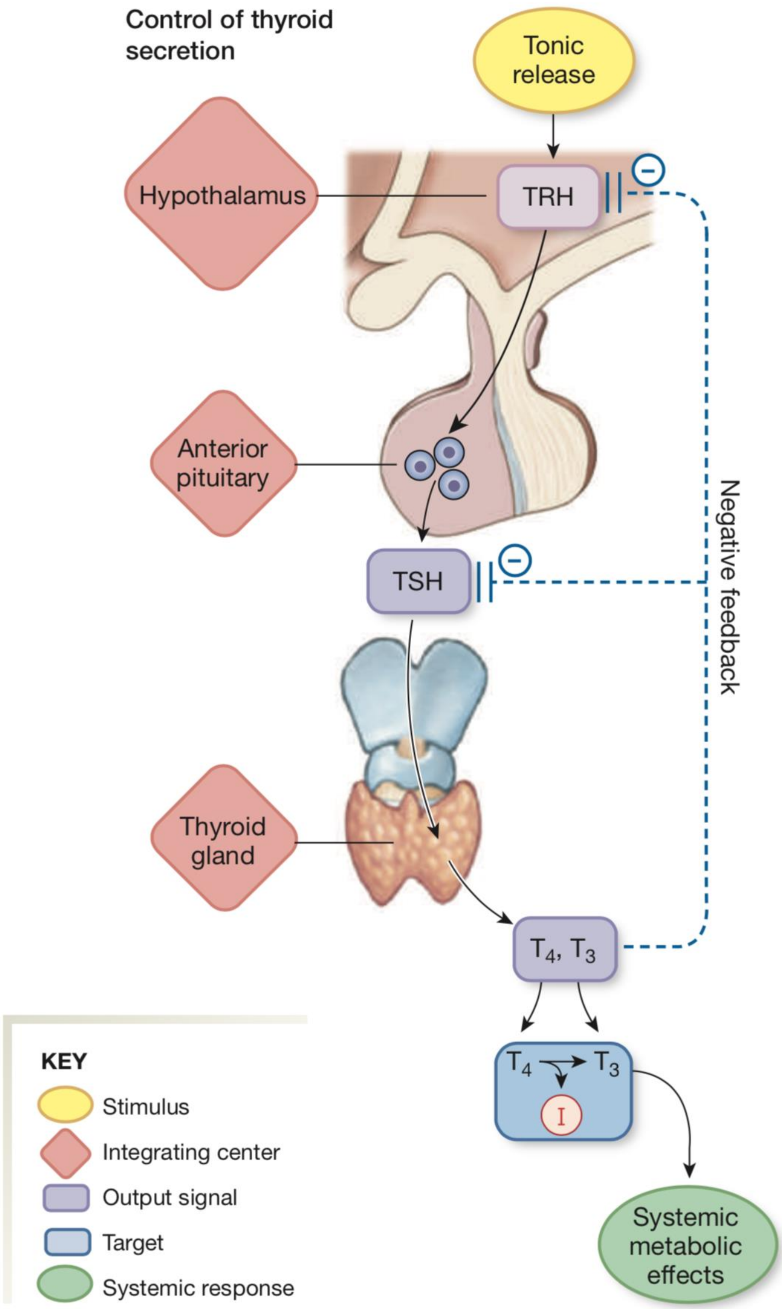
c. NFP = GHP + CHP + BCOP

d. NFP = GHP - CHP – BCOP

Questions 41 and 42 refer to the following information.

The control of thyroid hormone secretion follows the pathway depicted in the figure. Thyroid releasing hormone (TRH) is secreted by the hypothalamus, which stimulates the production of thyroid stimulating hormone (TSH) from the anterior pituitary. TSH then stimulates the production of the hormones thyroxine (T4) and triiodothyronine (T3) from the thyroid gland.

The thyroid gland requires dietary iodine to form T4 and T3. T4 contains four iodine atoms while T3 contains three.



41. Hypothyroidism can be caused by iodine deficiency. What is the likely consequence of hypothyroidism?

    a. High TSH, low T4 and low T3 levels.

    b. Low TSH, low T4 and low T3 levels.

    c. High TSH, low T4 and high T3 levels.

    d. Low TSH, high T4 and high T3 levels.

42. Graves' disease is an autoimmune disease where the body produces thyroid-stimulating immunoglobulins that activate TSH receptors on the thyroid gland. What is the likely consequence of Graves' disease?

    a. Decreased TRH, increased TSH, increased T4 and increased T3 levels

    b. Decreased TRH, decreased TSH, increased T4 and increased T3 levels

    c. Increased TRH, decreased TSH, decreased T4 and decreased T3 levels

    d. Increased TRH, increased TSH, increased T4 and increased T3 levels

43. John takes 1g of soil and adds 9mL of water (making total volume 10mL). Assume all the bacteria in the soil end up evenly distributed in this suspension. Two serial 10-fold dilutions of this soil suspension are performed and 0.2mL of the last dilution is plated on an agar plate. After a two-day incubation period, 250 colony forming units (CFU) were counted on the agar plate. What is the concentration of bacteria in the initial soil sample?

    a. 2.50 x 106 CFU/gram of soil

    b. 1.25 x 105 CFU/gram of soil

    c. 1.25 x 106 CFU/gram of soil

    d. 1.25 x 107 CFU/gram of soil

Question 44 refers to the following information

Bacteriophages, also known as phages, are viruses that can infect and kill bacteria. The susceptibility of a bacterial strain to infection by phages can be measured using a plaque assay.

The process of performing a plaque assay is: Phages are added to a lawn of bacterial cells growing on a plate. When one phage infects a cell, this infected cell is lysed and new phages are released. These newly released phages infect and lyse neighboring cells (the phages cannot travel to distant cells). The infection process proceeds until the area of cell lysis becomes so large that it is visible by the naked eye. This area of visible cell lysis, caused initially by one phage, is called a plaque. Therefore, the number of plaques is an estimate of the number of phages that were initially added to the lawn of bacterial cells. This is the plaque count.

Let the efficiency of plaquing (EOP) be the ratio of a mutant bacterial strain's plaque count to that of the wild type.

44. Using the above definition of EOP, which of the following is CORRECT?

    a. An EOP value equal to 1 is impossible.

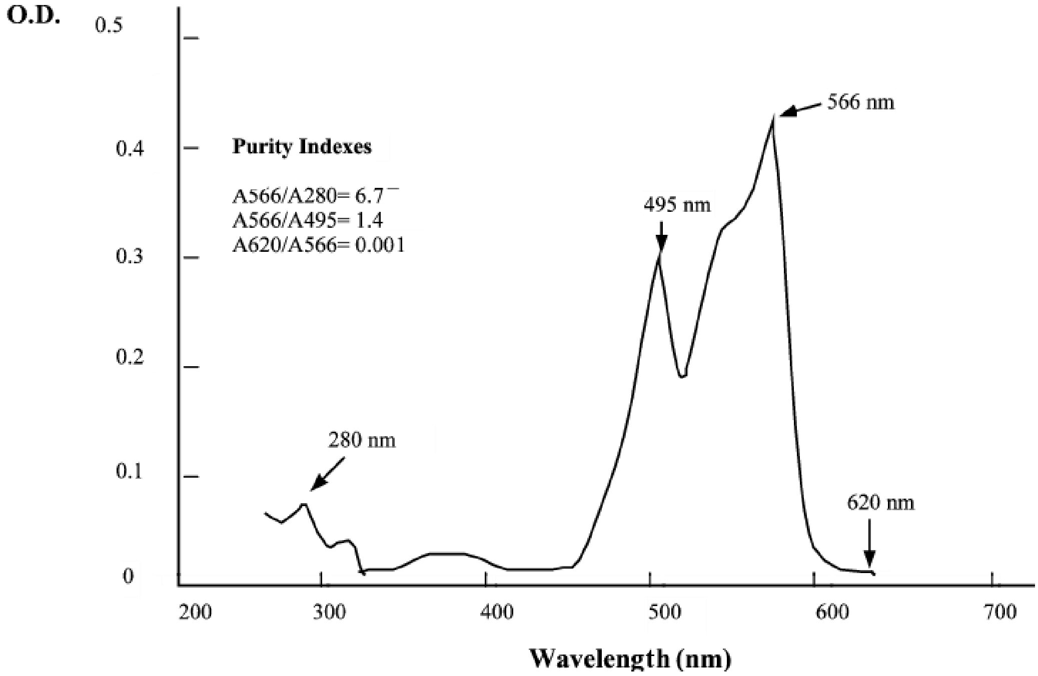
    b. An EOP value over 1 is impossible.

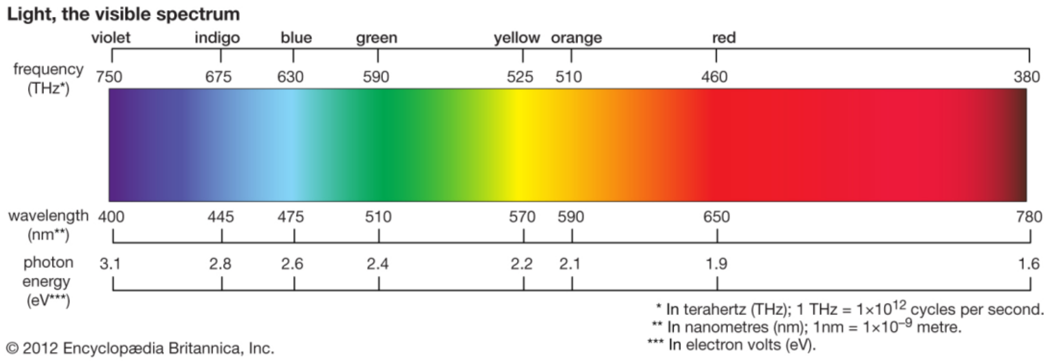
    c. An EOP value of 10-5 represents greater susceptibility to phage infection than a value of 10-1

    d. An EOP value of 10-5 represents greater resistance to phage infection than a value of 10-1

The following information relates to question 45

The absorption spectrum of a photosynthetic pigment and the spectrum of visible light are given below.





45. What is the most likely colour of this photosynthetic pigment?

a. Green

b. Red

c. Violet

d. Yellow

Questions 46 - 53 refer to the following information.

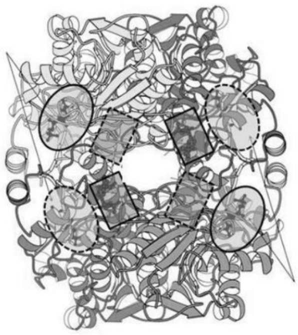
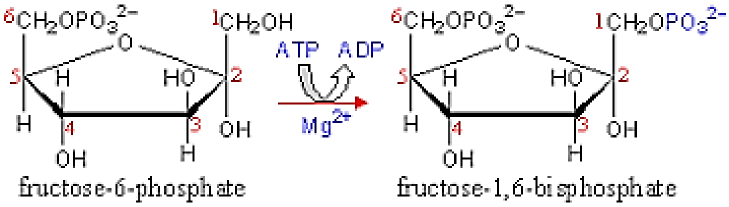


Figure 1. 3D visualisation of the protein backbone structure for an enzyme involved in glycolysis. Please note some of the following questions relate to the circles and rectangles positioned on this visualisation.

The reaction that this enzyme catalyses is:



The enzyme is composed of four subunits. There are certain sites on the enzyme that can bind to other molecules, indicated by the rectangles and ovals. Binding sites at the front of the molecule are indicated by solid lines. Binding sites at the back of the molecule is indicated by dotted lines. Oval sites can bind to fructose-6-phosphate. The sites indicated with rectangles bind fructose-2,6-bisphosphate. Interestingly, ATP can bind to both sites (ovals and rectangles), but ADP can only bind to rectangular sites.

ATP, ADP, and fructose-6-phosphate are all negative molecules.

In addition, please note that the term allosteric relates to enzymes that change the shape of the binding site of a molecule.

46. Enzymes catalyse biochemical reactions by altering which of the following quantities associated with the reaction?

   a. The enthalpy of formation, ΔH

   b. The equilibrium constant, Keq

   c. The change in Gibb’s free energy, ΔG

   d. The activation energy, Ea

47. Choose the best option to complete the sentence: The ovals indicate_______ and the rectangles indicate ________.

    a. allosteric sites, catalytic sites.

    b. catalytic sites, allosteric sites.

    c. where allosteric activators bind, where allosteric inhibitors bind.

    d. where allosteric inhibitors bind, where allosteric activators bind.

48. Because the molecules that bind to the sites indicated with both ovals and/rectangles are_________, you would expect that the amino acids lining these sites to include_________

    a. positively charged, Asp and Glu which are negatively charged.

    b. negatively charged, Lys and Arg which are positively charged.

    c. negatively charged, Asp and Glu which are negatively charged.

    d. positively charged, Lys and Arg which are positively charged.

49. Deficiency in the muscle isoform of the enzyme depicted in Figure 1 causes Tauri's disease (Type VII glycogen storage disease). The metabolic outcomes in muscle cells are likely to include:

    a. decreased fructose 1,6 bisphosphate, increased levels of fructose-6-phosphate

    b. decreased fructose 1,6 bisphosphate, decreased levels of fructose-6-phosphate

    c. increased fructose 1,6 bisphosphate, increased levels of fructose-6-phosphate

    d. increased fructose 1,6 bisphosphate, decreased levels of fructose-6-phosphate

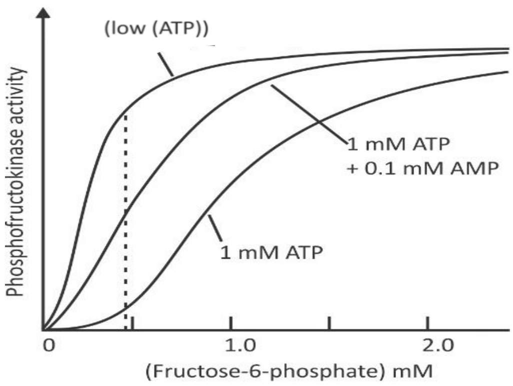


Fig 2A: Regulation of PFK activity

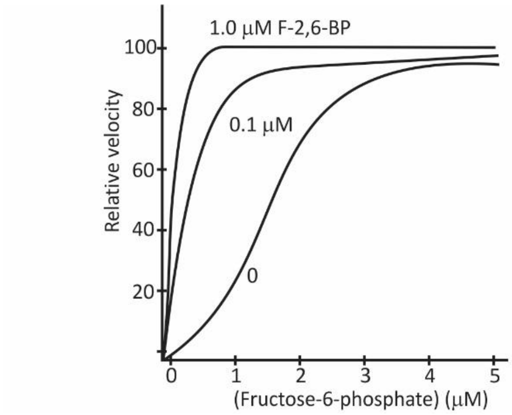


Fig 2B: Graph between the relative velocity and fructose-6 phosphate

Figure 2A and 2B: Enzyme activity under different conditions. (PFK = Phosphofructokinase)

50. Figure 2A shows that for this enzyme ATP is a/an:

    a. allosteric activator.

    b. allosteric inhibitor.

    c. product.

    d. substrate.

51. Figure 2A and Figure 2B show that when [fructose 6-phosphate] is high the enzyme:

    a. exhibits product inhibition.

    b. exhibits non-competitive inhibition.

    c. is stabilised in the high activity state.

    d. is stabilised in the low activity state.

52. The K_M of an enzymatic reaction is defined as the substrate concentration where there is enzymatic activity is half that of Vmax (maximum enzyme activity). Which of the following statements is correct?

    a. A high K_M is indicative of high enzyme efficiency.

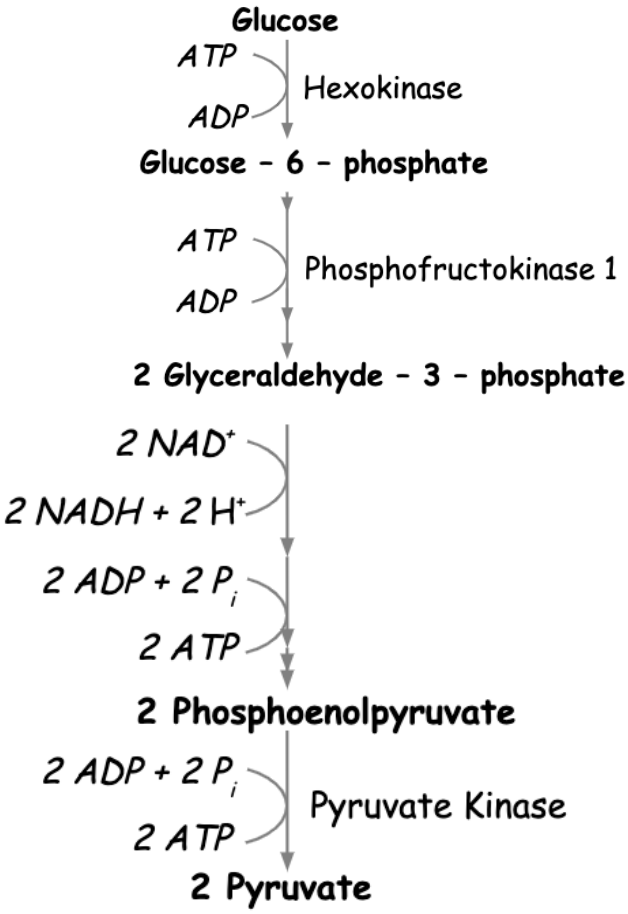
    b. A low K_M is indicative of a high enzyme efficiency.

    c. K_M is calculated by doubling Vmax.

    d. K_M is calculated by halving Vmax.

Questions 53 - 56 refer to the following information.

Cellular respiration is a complex process involving many steps. Generally, these steps are grouped into 3 main phases: glycolysis, the citric acid cycle, and the electron transport chain. The process of glycolysis can be summarised as follows:



53. Given this information, how many carbon atoms does a pyruvate molecule contain?

a. 1

b. 2

c. 3

d. 4

54. During glycolysis, there are steps at which ATP is created from ADP and phosphate and steps at which ATP is broken down. Which of the following options can be ruled out based on the information provided?

   a. The creation and breakdown of ATP occur at different locations within the cell.
   b. The order of reactions is irrelevant, net ATP produced is the same.
   c. The breakdown of ATP releases energy that is required for glycolysis to proceed.
   d. Glycolysis involves energy investment and payoff phases which are both necessary for respiration.

55. Glycolysis utilises a total of 10 enzymes, including those mentioned in the above diagram. How might the cell benefit from using many different enzymes in cellular respiration?

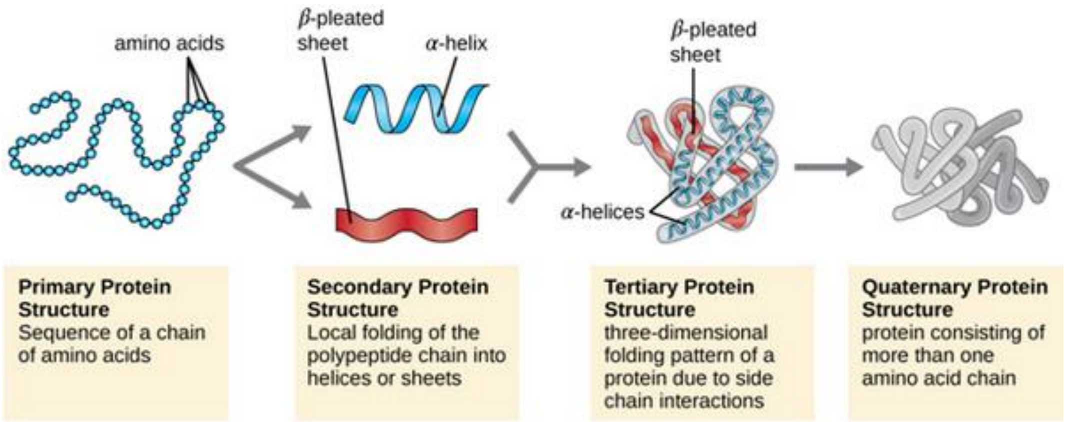
   a. Activation energy of a reaction is decreased, allowing it to occur more quickly.
   b. The cell needs fewer points at which it can promote function.
   c. Using specialised enzymes allows each step of respiration to occur more slowly.
   d. The cell needs fewer points at which it can limit enzyme function to control overall energy output.

56. Given the necessity of cellular respiration to life, which of the following features would you NOT expect?

   a. Negative feedback loops
   b. Positive feedback loops
   c. High levels of regulation
   d. Susceptibility to mutation

Questions 57 - 60 refer to the following information.

Review the following diagram and answer the 4 subsequent questions.



57. A protein's primary structure consists of a series of amino acids held together by peptide (covalent) bonds. Which of the following would alter the primary protein structure?

    a. Boiling the sample

    b. Increasing pH of the solution

    c. Decreasing pH of the 0solution

    d. Agitating the solution with a stirring rod

58. If a mutation occurred which caused a protein's primary structure to be half the original length, what is the most likely effect this would have on the cell?

    a. Mutated proteins would fold, resulting in impaired cellular function.

    b. The mutated amino acid chain would be hydrolysed, resulting in no net effect on the cell.

    c. The defect in the protein would spread to other nearby cells, forming a plaque of malformed proteins.

    d. The protein would still be able to work normally if it had a quaternary structure, since each individual polypeptide chain comprises only a small part of the overall protein.

59. A researcher is trying to synthesise a protein with tertiary structure. They have been able to produce a copy of the amino acid sequence, but have been unsuccessful in creating even one properly folded protein. What is the most likely reason for this?

a. The experiment's temperature is too low.

b. The solution is deficient in critical amino acids.

c. The researchers have isolated the wrong segment of DNA

d. The experiment's pH does not mimic the correct cellular conditions.

60. A prion is a rare type of misfolded protein that can trigger other correctly folded proteins of the same type to fold incorrectly.
Which of the following is LEAST likely to be an origin for prion disease in humans?

a. Unusually rapid cellular proliferation

b. Eating contaminated vegetables

c. Spontaneous mutation

d. Environmental factors leading to abnormal cellular conditions

END OF EXAM

SOURCES

Question 1 - 3: Image sourced: https://gentleforindustrial.blogspot.com/2018/06/animal-cell-blank-diagram.html

Question 7 - 14: Images sourced: http://academic.udayton.edu/ryanmcewan/Courses/Ecology/Exam2,PrepMaterials.pdf

Question 15 – 21: Adapted and modified from past IBO paper

Question 22: Image sourced and modified: https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential

Question 28: Image sourced: (Claudio Muheim 2017) https://www.semanticscholar.org/paper/Antibiotic-uptake-in-Gram-negative-bacteria-Muheim/eec44199c3299dd38172dd41fbfbaf0dcfd0e71f

Question 29: Image sourced and modified: (Benedetti et al., 2018)

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Question 36 – 37: Phases of the cell cycle image sourced: http://xaktly.com/CellCycle.html

Cytometry graph sourced and modified: Figure 17-8 Molecular Biology of the Cell (Alberts 6e),

Question 38: Image sourced and modified: (Groshel & Bushman 2005), Cell Cycle Arrest in G2/M Promotes Early Steps of Infection by Human Immunodeficiency Virus.

Question 39: Image sourced and modified: Khan Academy.

Questions 41 and 42: Image sourced and modified: Human Physiology: An Integrated Approach (7e) by Dee Silverthorn.

Question 45: Image sourced: (Rossano et al., 2003) Extracting and purifying R-phycoerythrin from Mediterranean red algae Corallina elongata

Question 46 – 49: Source: https://alchetron.com/Phosphofructokinase-1 AND http://biosiva.50webs.org/glycol.htm

Question 50 – 52: Image sourced: https://www.letstalkacademy.com/publication/read/regulation-of-glycolysis-regulation-of-pfk1

Question 57 – 60: Image sourced: http://easylifescienceworld.com/levels-of-organization-in-proteins/